

Jacob Zheng

☎ +1 613-219-2816 | ✉ jacob.zheng2007@gmail.com | 🔗 linkedin.com/in/zhe-jac | 🐙 github.com/zhe-jac

EDUCATION

University of Waterloo

Bachelor of Mathematics, Honours, Co-op

Waterloo, ON

Expected Graduation Apr 2030

- President's Scholarship Recipient

TECHNICAL SKILLS

Languages: TypeScript, Java, Python, CSS, JavaScript, C++, C, SQL

Frameworks / Libraries: Node.js, React, Next.js, TailwindCSS, FastAPI, NumPy, PyTorch

Developer Tools: Git, VS Code, Vercel, Cursor, Docker

PROJECTS

Multimodal AI Teaching Assistant / *React, Next.js, FastAPI, Python, ElevenLabs, Gemini API*

- Developed a **real-time agentic AI tutoring system** that identifies and analyzes student mistakes on a **TLDraw whiteboard** using the Gemini API, with ElevenLabs Conversational AI providing live spoken feedback at **under 75ms latency**.
- Integrated **MediaPipe Face Mesh** directly in the browser to monitor student engagement in real time, processing **468 facial landmarks** to detect signs of disengagement with **90% accuracy**.
- Orchestrated **three external APIs** (ElevenLabs Scribe, Gemini, TLDraw) within a **stateless session architecture**, implementing WebSocket reconnection logic and in-memory state management to keep all services reliably in sync.

Cross-Cultural Music Trends Analyzer / *React, TypeScript, Spotify API, FastAPI, PostgreSQL, D3.js*

- Built a **data pipeline** to collect and aggregate top chart data across **50+ countries** via the Spotify API, capturing millions of data points on regional listening patterns.
- Applied **clustering algorithms** to group nations by musical similarity, surfacing cross-cultural listening trends across audio features like BPM, key, and energy.
- Designed an **interactive world map dashboard** visualizing country-level music taste.
- Analyzed correlations between music taste clusters and **geographic, linguistic, and cultural factors** to uncover what drives regional listening habits.

EXPERIENCE

Ski Instructor

Camp Fortune

Dec 2022 – Jan 2025

Chelsea, QC

- Instructed **30+ students per day**, adapting to varied learning styles while maintaining safety standards and CSIA Level 1 Certification.

Court Attendant

Barrhaven Tennis Club

Sept 2022 – Aug 2025

Ottawa, ON

- Supported operations for a **250-member** facility, assisting **50+ members per shift** and facilitating bookings, events, and tournaments.